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CRC-Aided External Iterative LDPC Decoding Algorithm for Slow Frequency-Hopping Interference Scenarios

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***Abstract*—In short-burst slow frequency-hopping (SFH) gaussian minimum shift keying (GMSK) systems under strong partial-band interference (PBI), the slot-varying interference state causes severe mismatch of the input log-likelihood ratios (LLR) for new radio low-density parity-check (NR-LDPC) decoding when channel state information (CSI) is unavailable. To address this issue, this paper proposes a cyclic redundancy check (CRC)-aided external iterative interference estimation and decoding scheme. Instead of using CRC only for post-decoding error detection, CRC-validated code blocks are exploited to reconstruct reliable reference symbols, based on which a slot-level residual metric is formed to estimate the interference impact and adaptively re-weight the LLR in an external loop. The proposed approach effectively suppresses the propagation of high-confidence errors in the Tanner graph and accelerates the convergence of LDPC decoding. Simulation results under typical PBI scenarios demonstrate that the proposed scheme significantly improves the bit error rate (BER) performance and approaches the ideal CSI benchmark without introducing additional pilot overhead.**

Keywords—slow frequency-hopping (SFH), partial-band interference (PBI), NR-LDPC, CRC-aided decoding, external iteration, interference estimation, LLR reweighting

I. INTRODUCTION

Frequency-hopping spread spectrum (FHSS) communications have been widely adopted in military and anti-jamming wireless systems due to their strong frequency diversity and interference resilience. According to the relationship between the hopping rate and the symbol rate, frequency-hopping systems can be categorized into fast frequency-hopping (FFH) and SFH. While FFH provides strong anti-interference capability by performing multiple hops within one symbol period, it requires extremely fast frequency synthesizers and strict synchronization accuracy. In contrast, SFH systems allow multiple symbols to be transmitted within each hopping slot, which significantly reduces hardware complexity while still maintaining good anti-jamming performance [1]. Typical SFH communication systems employ short-burst transmissions with high hopping rates, where multiple symbols are transmitted within each hopping slot. A representative example is the Tactical Targeting Network Technology (TTNT) data link [2].

In SFH systems, when a hopping slot encounters strong PBI, the reliability of the corresponding LLR of multiple consecutive symbols may degrade simultaneously. Such burst interference can easily lead to decoding failures if the receiver cannot accurately characterize the interference state. Therefore, designing robust channel coding and interference mitigation techniques for SFH systems remains an important research problem.

Early SFH systems primarily relied on Reed–Solomon and convolutional codes to mitigate burst errors. With the development of modern communication systems, low-density parity-check (LDPC) codes have attracted increasing attention due to their near-Shannon-limit performance and efficient iterative decoding algorithms [3]. In particular, 5G New Radio (NR) adopts LDPC codes for data channel coding [4]. Unfortunately, in practical SFH anti-jamming environments, interference parameters are typically unknown and may vary rapidly across hopping slots, leading to severe mismatch between the assumed and actual LLR statistics.

To address this problem, joint estimation and decoding (JED) approaches based on iterative information exchange have been widely investigated. Kang and Stark proposed an external iterative channel estimation and decoding method for turbo-coded FH systems, demonstrating that iterative updates of soft metrics can significantly narrow the performance gap between unknown and known interference states [5]. Subsequent studies extended this idea to various coding structures and estimation strategies [6].

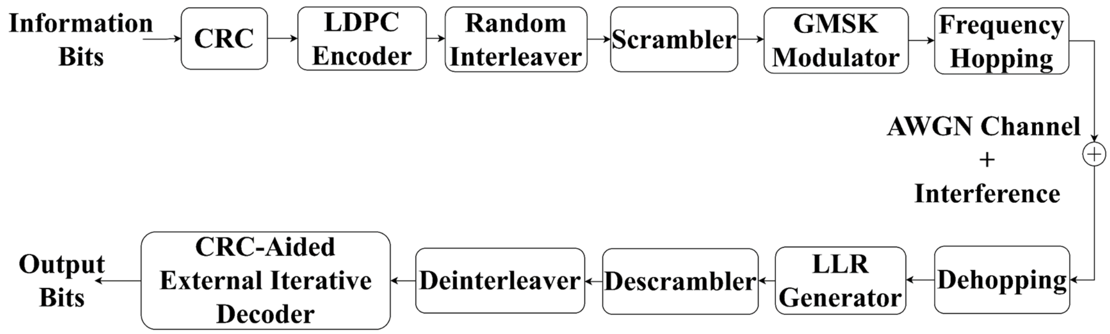

Fig. 1. System model.

In practical communication systems, CRC codes are widely employed for error detection at the transport block or code block level [7]. Although CRC is traditionally used only for post-decoding verification, it inherently provides highly reliable structural information about the transmitted data. Recent studies have shown that CRC information can also be exploited to assist decoding or improve reliability estimation in modern coding schemes [8]–[10]. However, the potential of CRC as a reliable reference for interference identification and channel state estimation in SFH anti-jamming systems has not been sufficiently explored.

Motivated by this observation, this paper proposes a CRC-aided external iterative interference estimation and decoding framework for NR-LDPC coded SFH GMSK communication systems. The proposed method utilizes CRC-validated code blocks to reconstruct reliable reference signals, enabling slot-level interference estimation and dynamic correction of LLR weights through an external iterative loop. Simulation results demonstrate that the proposed scheme effectively mitigates LLR mismatch under strong partial-band interference and significantly improves the decoding convergence performance without introducing additional pilot overhead.

II. System Model and Problem Analysis

A. System Model

Consider a short-burst slow frequency-hopping communication system as illustrated in Fig. 1. At the transmitter, the information bits are encoded using CRC and NR-LDPC coding, followed by interleaving and scrambling to mitigate burst interference. The encoded sequence is then modulated and transmitted over the frequency-hopping channel.

The encoded bit stream is modulated using GMSK combined with frequency hopping. The complex envelope of the transmitted signal in the k-th hopping slot can be expressed as

$$x_k(t) = \exp(\mathrm{j}\pi h \textstyle\sum_n a_k[n] q(t-nT))\, e^{\mathrm{j}2\pi \Delta f_k t} \tag{1}$$

In a strong interference environment, the received signal is affected not only by multipath fading and thermal noise, but also by the superposition of external interference. After dehopping, the baseband equivalent received signal in the k-th hopping slot can be expressed as

$$r_k(t) = x_k(t) e^{-\mathrm{j}2\pi \Delta f_k t} + i_k(t) + n(t) \tag{2}$$

where $n(t)$denotes complex Gaussian noise and   represents the interference component. After differential demodulation, the initial LLR are calculated from the differential phase observations and used as the input of the LDPC decoder.

B. Interference Characteristics and LLR Mismatch Analysis

In a SFH system, since the dwell time T_{hop} is much larger than the symbol period T_{sym}, the channel exhibits typical block fading characteristics. An NR-LDPC encoded codeword is mapped to k independent hopping slots after interleaving. When partial frequency points encounter strong interference, the interference displays significant binary state distribution characteristics. Assuming the interference bandwidth coverage probability is ρ , we introduce an interference indicator variable$S_k \in \{0,1\}$. At this point, equivalent noise variance $\sigma_{\phi,k}^2$ in the k-th slot can be modeled as

$$\sigma_{\phi,k}^2 = (1-S_k)\sigma_0^2 + S_k\sigma_1^2 \tag{3}$$

where σ_0^2 represents the background thermal noise variance, and $\sigma_1^2 = \sigma_0^2 + \sigma_J^2$ is the total variance under interference, typically satisfying $\sigma_1^2 \gg \sigma_0^2$. The performance of the NR-LDPC decoder is highly dependent on the accuracy of the input LLR. Under ideal conditions, assuming the phase noise follows a Gaussian distribution $\mathcal{N}(0, \sigma_{\phi,k}^2)$, the ideal LLR and the differential phase $\Delta\phi_k[n]$exhibit a linear relationship. For a binary input channel, the log-likelihood ratio is defined as

$$L(n) = \log\frac{p(y_n|b_n=0)}{p(y_n|b_n=1)} = \frac{2\mu}{\sigma_{\phi,k}^2}\Delta\phi_k[n] \tag{4}$$

Equation (4) indicates that optimal LLR calculation requires dynamic adjustment of the slope $2\mu/\sigma_{\phi,k}^2$ based on the instantaneous signal-to-noise ratio (SNR) of each frame. However, in short-burst frequency-hopping systems, the extremely short dwell time makes it difficult for the receiver to accumulate sufficient statistical samples for real-time and accurate estimation of the instantaneous noise variance $\sigma_{\phi,k}^2$. Consequently, baseline receivers typically employ a fixed scaling factor κ to calculate soft information i.e.,$\widetilde{L_k}[n] = \kappa\Delta\phi_k[n]$.

This fixed weighting strategy implicitly assumes that the channel noise level remains constant at $\sigma_{\phi,0}^2$. When the system encounters strong interference i.e., $\sigma_{\phi,k}^2 \gg \sigma_{\phi,0}^2$, the absolute value of the true LLR should be significantly reduced to reflect the low reliability. However, the fixed coefficient κ leads to an excessive LLR magnitude. This phenomenon causes the decoder to assign overly high confidence to erroneous information originating from interfered frames. Such high-confidence errors propagate rapidly through the LDPC Tanner graph, violating parity-check constraints and leading to the divergence of iterative decoding.

In a communication environment where interference and noise coexist, it is assumed that the equivalent channel after differential demodulation and deinterleaving can be approximated as a memoryless Binary Input Additive White Gaussian Noise (BI-AWGN) channel. Let the input be $X \in \{+1, -1\}$ and the received signal be $Y = X + N$, where $N \sim \mathcal{N}(0, \sigma^2)$. In this case, the channel capacity $C(\gamma)$ depends solely on the signal-to-noise ratio (SNR) $\gamma = E_b/N_0$, which can be expressed as

$$C(\gamma) = 1 - E_{Z\sim\mathcal{N}(0,1)}\left[\log_2\left(1 + e^{-2\sqrt{\gamma}Z-2\gamma}\right)\right] \quad (5)$$

For partial-band interference channels, the system operates in a two-state mixture mode: it remains in an interference-free state with probability $1 - \rho$ and enters an interfered state with probability ρ. Assuming the receiver possesses perfect CSI, the ergodic capacity of this mixture channel is the weighted sum of the capacities of the two states:

$$C_1 = (1 - \rho)C(\gamma_0) + \rho C(\gamma_1). \quad (6)$$

This formula provides the maximum information transmission rate theoretically achievable by the system when the LLR is perfectly matched and the codeword length tends toward infinity. Due to the short-burst transmission adopted in this system, the length of each coding block is finite, making the asymptotic capacity unattainable. According to the finite blocklength information theory proposed by Polyanskiy et al. [11], the approximate bound of the maximum achievable code rate R under the constraints of block length E and target block error rate ϵ is expressed as

$$R \approx C_1 - \sqrt{\frac{V}{E}}Q^{-1}(\epsilon) + \frac{\log_2 E}{2E} \quad (7)$$

where V is the information density variance, and $Q^{-1}(\cdot)$ is the inverse cumulative distribution function of the standard normal distribution. It is observed that even under the dual constraints of strong interference and short blocklength, there still exists a clear capacity-approaching target for NR-LDPC codes in theory. If the LLR bias can be effectively corrected, the system will regain the capability to approach these theoretical limits.

Therefore, in SFH systems with strong interference, the main performance degradation originates from the mismatch between the assumed and actual LLR statistics. To address this problem, a CRC-aided residual-driven external iterative decoding algorithm is proposed in the next section to estimate the slot-level reliability and adaptively correct the LLR weights.

III. Proposed CRC-Aided External Iterative Decoding Algorithm

In the proposed receiver, CRC information is incorporated into an external iterative decoding framework. After initial demodulation and LLR generation, the LLR sequence is divided according to the hopping slots. The LDPC decoder performs iterative decoding, and CRC results are used to identify reliable code blocks for subsequent interference estimation and LLR correction.

Due to the time-varying interference in SFH channels, accurate channel state information is difficult to obtain at the receiver. Therefore, a CRC-aided residual metric is introduced to estimate the interference level of each hopping slot and iteratively refine the LLR weights. The detailed procedure of the proposed CRC-aided weighting algorithm is described as follows.

After the *t*-th external iteration, the CRC indicator $c_b^{(t)}$ is obtained for the *b*-th code block. When $c_b^{(t)} = 0$, the corresponding code block is considered reliable. This block then undergoes NR-LDPC re-encoding and rate-matching to generate a reconstructed codeword stream $\nu(t)$. Simultaneously, a reliability mask $m^{(t)}$ is constructed to mark the bit positions contributed by the CRC-validated blocks. Let $\pi(\cdot)$ denote the random interleaving permutation; the reconstructed stream and mask in the interleaving domain are represented as $\widehat{\nu^{(t)}} = \pi(\nu(t))$ and $\widehat{m^{(t)}} = \pi(m^{(t)})$, respectively. Let $X_{k,n} \in \{+1, -1\}$ denote the transmitted coded symbol at the *n*-th position of the *k*-th hopping slot, and let $L_{k,n}^{(0)}$ denote the initial descrambled LLR. For a binary-input channel, the posterior probabilities implied by $L_{k,n}^{(0)}$ are

$$\begin{aligned} P\left(X_{k,n} = +1 \mid L_{k,n}^{(0)}\right) &= \frac{e^{\{L_{k,n}^{(0)}\}}}{1+e^{\{L_{k,n}^{(0)}\}}}, \\ P\left(X_{k,n} = -1 \mid L_{k,n}^{(0)}\right) &= \frac{1}{1+e^{\{L_{k,n}^{(0)}\}}} \end{aligned} \quad (8)$$

Therefore, the posterior mean soft symbol can be written as

$$\begin{aligned} \overline{x}_{k,n}^{(0)} &= E[X_{k,n} | L_{k,n}^{(0)}] \\ &= P(X_{k,n} = +1 | L_{k,n}^{(0)}) - P(X_{k,n} = -1 | L_{k,n}^{(0)}) \\ &= \tanh\left(\frac{L_{k,n}^{(0)}}{2}\right) \end{aligned} \quad (9)$$

which represents the soft symbol expectation under the BI-AWGN assumption.

The reconstructed bits are mapped to equivalent baseband symbols as $\widehat{x_{k,n}^{(t)}} = 1 - 2\widehat{v_{k,n}^{(t)}} \in \{+1, -1\}$,The bit-wise residual is then calculated as $e_{k,n}^{(t)} = \left(\overline{x}_{k,n}^{(0)} - \widehat{x_{k,n}^{(t)}}\right)^2$ which measures the mismatch between the LLR-implied soft symbol expectation and the reconstructed symbol.

Finally, the average residual for the *k*-th hopping slot is defined as

$$\begin{aligned} E_k^{(t)} &= \frac{1}{|S_k^{(t)}|}\sum_{n\in S_k^{(t)}} e_{k,n}^{(t)} \\ &= \frac{1}{|S_k^{(t)}|}\sum_{n\in S_k^{(t)}} \left(\bar{x}_{k,n}^{(0)} - \hat{x}_{k,n}^{(t)}\right)^2 \end{aligned} \quad (10)$$

where $S_k^{(t)}$ is the set of reliable bit positions in the *k*-th slot, defined by all bit positions validated by the CRC. From a statistical perspective, $E_k^{(t)}$ reflects the average reliability mismatch of the k-th hopping slot estimated from CRC-validated bits. A larger residual indicates stronger interference or more severe LLR mismatch in the corresponding slot. To avoid estimation jitter caused by insufficient samples, the weight of a slot is not updated when $|S_k^{(t)}| < N_{\min}$. To improve robustness against outliers caused by strongly interfered slots, we choose a reference residual $\gamma_E^{(t)}$ as the median of $\{E_k^{(t)}\}$. Equivalently, $\gamma_E^{(t)}$ can be obtained by minimizing the l_1 deviation:

$$\gamma_E^{(t)} = \arg\min_{\gamma} \sum_{k:|S_k^{(t)}|\geq N_{min}} |E_k^{(t)} - \gamma| \tag{11}$$

In an ideal receiver, the LLR of the n-th coded symbol in the k-th hopping slot should be scaled according to the slot-dependent effective noise variance. Under the equivalent BI-AWGN assumption, the matched LLR can be expressed as

$$L_{k,n}^{*} = \frac{2r_{k,n}}{\sigma_k^2}, \tag{12}$$

where $r_{k,n}$ denotes the corresponding soft observation and σ_k^2 represents the effective noise variance of the k-th hopping slot. In contrast, the baseline receiver computes the initial LLR using a fixed reference variance σ_{ref}^2, which leads to

$$L_{k,n}^{(0)} = \frac{2r_{k,n}}{\sigma_{\text{ref}}^2} \tag{13}$$

Therefore, the ideally matched LLR should satisfy $L_{k,n}^{\star} = \omega_k^{\star} L_{k,n}^{(0)}$, where $\omega_k^{*} = \sigma_{\text{ref}}^2/\sigma_k^2$ denotes the ideal slot-dependent scaling factor. Since the instantaneous slot variance σ_k^2 is unavailable in practice, the slot-level residual metric $E_k^{(t)}$ is employed as a surrogate indicator of the reliability mismatch in the k-th slot. A larger residual generally corresponds to a stronger interference level and thus requires a smaller LLR scaling factor. Since the residual $E_k^{(t)}$ reflects the mismatch between the LLR-implied soft symbol and the reconstructed reliable symbol, it is statistically correlated with the effective noise variance of the slot. Consequently, the slot weight is chosen to be inversely proportional to the residual metric.

To obtain a robust normalization reference, the median residual $\gamma_E^{(t)}$ defined in (11) is adopted. The resulting residual-driven weight is therefore defined as

$$\widetilde{\omega_k^{(t)}} = \max\left(\omega_{min}, \min\left(1, \frac{\gamma_E^{(t)}}{E_k^{(t)}+\epsilon}\right)\right) \tag{14}$$

where ϵ is a small positive constant to avoid numerical instability and $\omega_{\min}$ prevents excessively small weights for heavily interfered slots.

To suppress oscillations during external iterations, a damping coefficient $\lambda \in (0,1)$ is introduced, and the updated weight is given by

$$\omega_k^{(t)} = \lambda\omega_k^{(t-1)} + (1-\lambda)\widetilde{\omega_k^{(t)}} \tag{15}$$

Accordingly, the LLR used in the next external iteration is updated as

$$L_{k,n}^{(t+1)} = \omega_k^{(t)} L_{k,n}^{(0)}. \tag{16}$$

To clarify the execution logic of the proposed CRC-aided scheme, the external iteration procedure is summarized as follows. Let N_{out} denote the maximum number of external iterations. For each iteration $t \in \{1,2,\dots,N_{out}\}$:

1) Weighting and Decoding: Apply the current frame weights $\omega_k^{(t-1)}$ to the deinterleaved LLRs. Perform LDPC decoding and subsequent CRC verification for all code blocks.

2) Early Termination Check: If all code blocks pass the CRC check, the decoding process is considered completely successful. The external iteration is immediately terminated to save computational complexity.

3) Reference Reconstruction: If not all blocks pass, collect the code blocks that successfully passed the CRC. Re-encode these reliable blocks to reconstruct the reference symbols $x_{k,n}^{(t)}$.

4) Residual Estimation and Weight Update: Compute the slot-level residual metric using the reconstructed symbols, update the LLR weights $\omega_k^{(t)}$, and proceed to the next iteration.

IV. Simulation Results and Analysis

To evaluate the proposed scheme, a short-burst slow frequency hopping GMSK communication system is simulated in MATLAB under partial-band interference. The channel coding follows the 5G NR-LDPC standard with Base Graph 2 defined in 3GPP TS 38.212 [7]. The information block length is 1000 bits and a 24-bit CRC is appended, resulting in an effective coding rate R = 1/3.GMSK modulation is adopted with Gaussian filter parameter BT = 0.3 and modulation index h = 0.5. The number of samples per symbol is set to 8 and the system sampling rate is F_s = 8MHz. Each transmitted burst consists of 340 bits, including a 16-bit preamble, a 26-bit synchronization sequence SyncA, a 256-bit payload, a 26-bit synchronization sequence SyncB, and a 16-bit postamble.Slow frequency hopping is performed over N_f = 20 candidate carrier frequencies. The hopping sequence is generated by a PN generator of order m = 10 with polynomial $x^{10} + x^7 + 1$.

Partial-band interference is modeled by injecting complex Gaussian jamming signals on a subset of the hopping frequencies. The signal-to-jammer ratio (SJR) controls the interference power, while the interference ratio determines the fraction of jammed hopping frequencies. Since the noise power is significantly lower than the interference power in the considered scenarios, the performance degradation is dominated by partial-band interference. Therefore, the impact of AWGN can be regarded as negligible compared with the interference.

A. Decoding Convergence Analysis

Fig. 2 shows the BER convergence versus the number of LDPC internal iterations under partial-band interference, where the interference ratio is ρ = 0.4. In the simulation, two interference levels are considered with signal-to-jammer ratio SJR = 2 dB and SJR = 0 dB. Without external iteration, the BER decreases slowly as the number of internal iterations increases and eventually saturates at a relatively high level. This phenomenon becomes more evident under stronger interference when SJR = 0 dB. The result indicates that the receiver suffers from severe LLR mismatch in interfered hopping slots, which leads to persistent error propagation during LDPC decoding. After introducing the proposed CRC-aided external iteration, the

BER curves shift downward and the decoding convergence becomes significantly faster. Even with only one or two external iterations, the BER is greatly reduced compared with the case without external iteration, and the improvement becomes more pronounced under stronger interference.

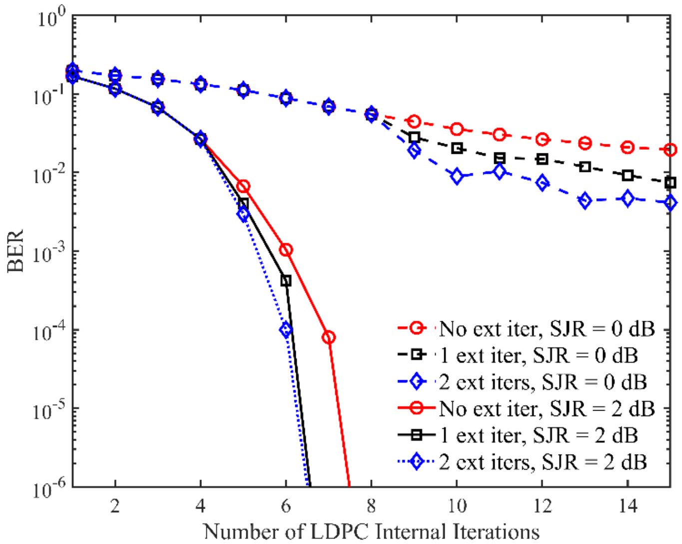

Fig. 2. Impact of external iterations on LDPC decoding performance under partial-band interference.

B. BER Performance Comparison

Building upon the decoding convergence analysis in Fig. 2, Fig. 3 illustrates the BER performance versus SJR under a dense interference scenario where ρ = 0.8.To ensure sufficient convergence during these evaluations, the NR-LDPC decoder is configured to perform a maximum of 30 internal iterations per external loop. The efficacy of the proposed strategy is further validated in the moderate interference scenario where $\rho = 0.4$, as shown in Fig. 4. Here, the receiver performance is compared against the Ideal CSI benchmark to assess estimation accuracy. At a target BER of 10^{-3}, a single external iteration achieves a performance gain of approximately 0.3 dB over the baseline. As the external iterations increase to three, the gain extends to 0.8 dB, with the BER curve nearly overlapping the Ideal CSI curve. This rapid convergence indicates that the residual-driven metric $E_k^{(t)}$ serves as an excellent surrogate for the unavailable instantaneous noise variance.

Fig. 5 compares the BER performance of the proposed CRC-aided external iterative scheme with 3 external iterations against the traditional soft-information-based iterative method [6] under varying SJR. For a fair comparison under the jamming ratio of $\rho = 0.4$, both methods employ 30 LDPC internal iterations. As observed, the proposed scheme consistently outperforms the traditional method across the evaluated SJR range. Specifically, at a target BER of 10^{-3}, the proposed scheme achieves a performance gain of approximately 0.2 dB over the traditional method. This superiority can be attributed to the reliability of the interference estimation source. The traditional method relies on soft metrics, which suffer from statistical inaccuracy due to the short-burst nature of the transmission.

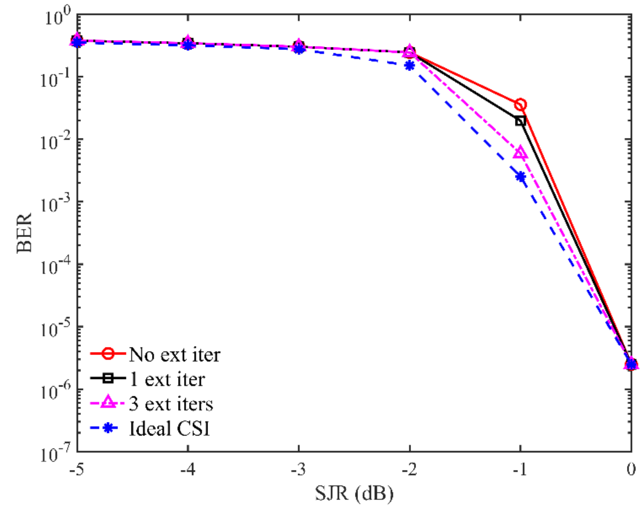

Fig. 3. Performance comparison of different receiving strategies under partial-band interference ($\rho = 0.8$) .

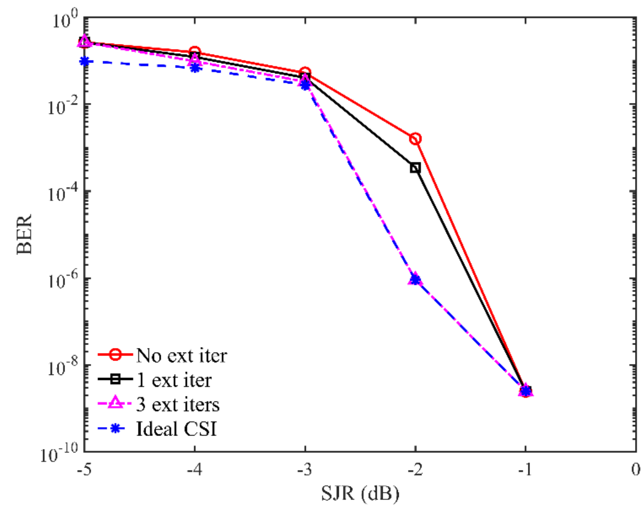

Fig. 4. Performance comparison of different receiving strategies under partial-band interference ($\rho = 0.4$).

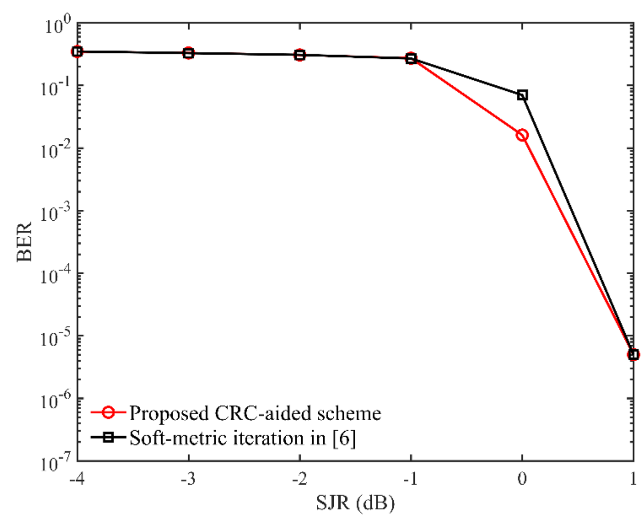

Fig. 5. Performance comparison between the CRC-aided external iteration and traditional soft information iteration in the SFH system.

In contrast, the proposed algorithm utilizes CRC-validated blocks as reliable references to precisely estimate interference parameters at the slot level.

C. Robustness under Dynamic Swept Jamming

To further evaluate the robustness of the proposed scheme in highly dynamic tactical environments, a continuous swept jamming scenario is introduced. We define α as the instantaneous fractional bandwidth of the swept jammer. Fig. 6 illustrates the bit error rate performance versus signal-to-jammer ratio under the conditions of α = 0.4 and α = 0.8.As observed from the simulation results, the conventional receiver suffers obvious performance degradation under both interference densities due to the rapid fluctuations of the slot states caused by the shifting interference band. Particularly under the heavy suppression environment of α = 0.8, the bit error rate curve of the conventional receiver almost stalls, making it difficult to improve performance simply by increasing signal power. In contrast, the proposed scheme exhibits good dynamic adaptability. Even with a single external iteration, the system can reasonably track the trajectory of the swept jamming across different slots through the CRC-aided residual estimation logic. When the number of external iterations increases to 3, the proposed scheme maintains robust decoding convergence and achieves considerable performance gains under both α = 0.4 and α = 0.8 scenarios. This indicates that the proposed algorithm can utilize fine-grained slot-level reliability evaluations to better adapt to time-varying interference patterns, thereby improving the transmission reliability of short-burst communication systems in dynamic electromagnetic environments.

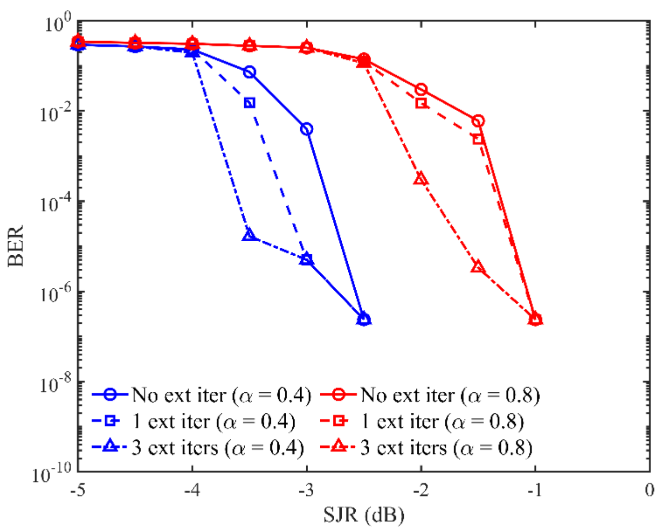

Fig. 6. BER versus SJR under continuous swept jamming for α = 0.4 and α = 0.8.

D. Complexity and Latency Analysis

To evaluate the complexity-performance trade-off, Fig. 7 illustrates the performance gain versus external iterations. The gain monotonically increases and saturates at 4 to 5 iterations. Building upon this convergence behavior, the proposed scheme's complexity is analyzed as follows:

1) Resource Consumption: The conventional receiver's complexity is heavily dominated by the internal NR-LDPC belief propagation decoding, which is proportional to $N_{in} \cdot E \cdot d_c$, where N_{in} is the number of inner iterations, E is the rate-matched block length, and d_c is the average check node degree. In contrast, the proposed external loop only adds lightweight CRC verification, element-wise residual calculations, and selective LDPC re-encoding strictly for CRC-validated blocks. This sparse re-encoding yields a minimal complexity proportional to E. Therefore, the additional hardware burden is marginal compared to the intensive internal LDPC node updates.

2) Processing Latency: Although external iterations increase worst-case decoding time, the proposed scheme employs an efficient early-termination mechanism that stops the loop immediately once all blocks pass the CRC validation. Under favorable conditions, decoding finishes in the first iteration, adding zero latency. Even under severe partial-band interference, as shown in Fig. 7, convergence rapidly saturates within 4 to 5 iterations. By capping the external loops at this threshold, the worst-case latency is strictly bounded, ensuringthe receiver satisfies the real-time constraints of short burst tactical communications.

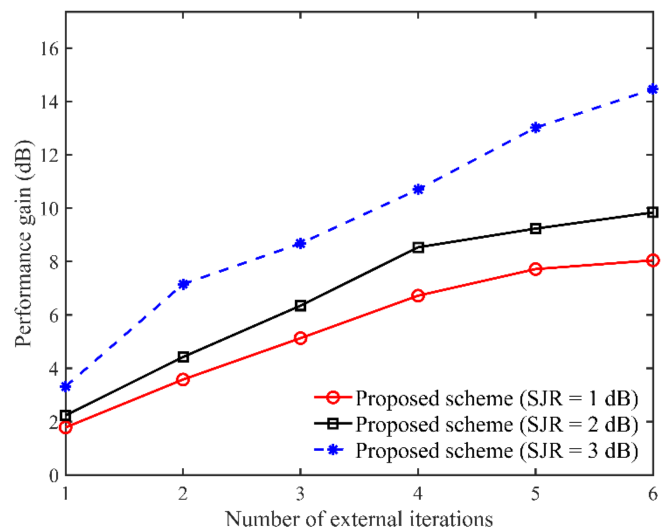

Fig. 7. Convergence characteristics of the performance gain of the CRC-aided external iteration with the number of iterations

V. Conclusion

This study proposes an anti-jamming decoding framework with zero additional pilot overhead for short-burst SFH systems. By repurposing CRC validation information to dynamically calibrate LLR weights, the proposed scheme achieves rapid convergence in strong interference environments without ideal CSI. Quantitative evaluations show that it outperforms traditional soft-metric methods by 0.2 dB and approaches the ideal CSI bound within just 4 to 5 external iterations, striking an optimal balance between anti-jamming capability and system complexity. Overall, this scheme serves as a useful reference and offers a new perspective for robust tactical communication designs.

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